

## FM12

### Open Source 12-Operator PM/FM Synthesizer

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FM12 is an easy-to-use 12-operator PM/FM synthesizer featuring a flexible modulation matrix and integrated stereo chorus. It is built in JUCE/C++ and the source code is bundled with this download, mostly written with assistance by Claude AI, and it is free and open source.

In FM, you have operators that can either be carriers or modulators. In classic FM, they are all sine waves. FM12 uses sine waves as default too, but you can also import your own single cycle waveforms (512, 1024 or 2048 samples). An operator acts as a carrier (audio output) when its matrix row is empty. When it has any route enabled in its row, it becomes a modulator, shaping the timbre of the operators it connects to.

### Default View

By default, FM12 shows all 12 Operators row by row. You can also reduce the amount of displayed Operators to keep the interface focused and immediately readable. Use the VIEW OP selector in the top bar to switch to any individual operator, or a subset of them (for example one, two, four or six at once). The single- or few-operator view makes it easy to understand what each operator is doing before expanding to the full patch view.

### Operator Parameters

Each of the 12 operators has the following controls:

| Control  | Parameter            | Description                                                                                                                                                  |
|----------|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DELAY    | Delay time           | Offset in time before this operator starts the sound output per played note.                                                                                 |
| ATT      | Attack time          | Duration for the operator to reach its peak level after a note starts.                                                                                       |
| DEC      | Decay time           | Duration to drop from the peak level down to the sustain level.                                                                                              |
| SUS      | Sustain level        | The amplitude held for as long as a key is pressed.                                                                                                          |
| REL      | Release time         | Duration to fade to silence after the key is released.                                                                                                       |
| VOL      | Operator level       | Output volume for carriers; modulation depth for modulators.                                                                                                 |
| RATIO    | Frequency multiplier | Multiplies the base note frequency to produce overtones.                                                                                                     |
| CNCL     | Cancel Dry Sound     | Cancels the dry sound of a modulated carrier so you only hear the effect the modulator has on the carrier, not the carrier itself anymore.                   |
| WAVEFORM | Waveform Selector    | Here you can choose between Sine, Triangle, Saw or Square Wave, but you can also import any 512-, 1024- or 2048 point single cycle waveform as sound source. |

### Controls & Features

| Control                                    | Description       | Details                                                                                                                                                                                 |
|--------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MATRIX                                     | Operator routing  | Grid connecting operator outputs (rows) to inputs (columns). An empty row means the operator is a carrier, it outputs sound directly. Otherwise, it will feed into the column input(s). |
| LOCKING (Right Mouse Click for Ratio Knob) | Ratio snap toggle | Per-operator toggle. When you right click and drag the ratio knob, the ratio values snap to multiples of 0.25 for clean harmonic intervals.                                             |
| FEEDBACK                                   | Self-modulation   | Diagonal knob for each operator. Allows an operator to modulate itself for added harmonic complexity. Two modes are available (see EXP FB toggle).                                      |
| CHORUS                                     | Amount & Width    | Controls the intensity and stereo spread of the built-in ensemble effect. Two knobs: Amount and Width.                                                                                  |

|                |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NYQUIST        | Anti-alias cutoff       | Sets a virtual Nyquist limit for frequency folding. Lowering this tames high-frequency aliasing at the cost of some brightness. Range: 100 Hz - 24000 Hz (covers 44.1 and 48 kHz sessions).                                                                                                                                                                                                                                                  |
| RANDOMIZE      | Matrix randomization    | Randomizes modulation assignments and operator volumes for operators 2-12. OP1 is always kept as a carrier.                                                                                                                                                                                                                                                                                                                                  |
| STAB           | Stab generator          | Generates a random stab patch with one of three modes (equal probability each): Classic: OP1 as carrier, modulators drawn from OP2-OP5. OP6 carrier: OP6 as carrier, OP7-OP12 as modulators with metallic inharmonic ratios and three routing flavours (Stack / Cluster / Hybrid). Both simultaneously: both carriers active at once. In this mode the OP6 modulators have a 50% chance of using harmonic ratios instead of inharmonic ones. |
| HALVE VOL      | Halve modulator volumes | Checks which operators currently have outgoing routes (i.e. are modulators) and halves their VOL. Minimum floor: 0.001. Press repeatedly to quietly fade modulation depth. This keeps the sound character, but makes it less harsh, great for fine tuning stabs generated with the STB button.                                                                                                                                               |
| FM ENGINE MODE | FM algorithm selector   | Selects the FM synthesis algorithm: Phase Modulation Mode (classic DX7-style), Linear FM Mode, Lin. FM Through Zero, Exponential FM Mode, or Exp. FM Through Zero. The "Through-Zero" Modes sound more like the classic FM sounds you know, the pure FM modes sound like radio sweeps.                                                                                                                                                       |
| EXP FB         | Experimental feedback   | Activates a delay-line feedback mode for the diagonal feedback knobs, which can turn a sine wave into a saw wave and beyond.                                                                                                                                                                                                                                                                                                                 |
| VIEW OP        | Operator view selector  | Selects which operator is shown in single-operator view (default: OP 1), or All to display all 12 operators simultaneously.                                                                                                                                                                                                                                                                                                                  |
| SAVE / LOAD    | Preset management       | Saves or loads a .fm12preset file anywhere on disk.                                                                                                                                                                                                                                                                                                                                                                                          |

### Quick Start

1. Load a preset with LOAD, or hit RND / STB for an instant patch.
2. Use VIEW OP to switch between operators, or select All to see the full matrix.
3. Use the MATRIX to route modulators. Empty rows = carriers.
4. Change VOL on modulators to control brightness and FM depth.
5. Press /2 to quietly dim modulation without changing the routing if the patch sounds too harsh.
6. Use FM ENGINE MODE to switch between PM, Linear FM, and Exponential FM flavours.
7. Add stereo width with the CHORUS knobs.
8. Lower NYQUIST for lo-fi aliasing effects (works correctly at 44.1 kHz and 48 kHz).

### Notes

*CPU note: FM12 uses a sine wave lookup table for efficiency, but 12-operator polyphony can still be demanding. Reduce voices if needed. Preset format: .fm12preset files are standard JUCE XML state files and can be opened in any text editor.*

Thanks for trying it out - have fun!

### License

FM12 is released under Creative Commons Attribution NonCommercial v3.

This means you can freely use the source code in your own, even commercial projects, but if you don't alter it to the point that it fully becomes your own creation, please keep your project open source too.